

**Nuclear Talent Course on Nuclear Theory for Nuclear Structure Experiments, ECT*,
July 3-21, 2017**

fig1.jpg

The present document aims at giving a summary, with background information as well, of the twelfth Nuclear Talent course, *Nuclear Theory for Nuclear Structure Experiments*. The course was held at the premises of the European Center for Theoretical Studies in Nuclear Physics and Related Areas (ECT*), Trento, Italy during the period July 3 to July 21, 2017.

This was the fourth course which has been held at the ECT*, the first one in 2012, then 2014 and 2015 and finally this one in 2017. The support from the ECT* has simply been crucial for the successful arrangement and running of these courses. In total 52 applications were received and a final of 31 participants were selected. Five teachers were involved, with two of them being the organizers.

I. REPORT

A. Introduction

B. Aims and outcomes

We held approximately forty-five hours of lectures over three weeks and a comparable amount of practical computer and exercise sessions, including the setting of individual problems and the organization of exercises and projects.

The mornings consisted of lectures and the afternoons was devoted to exercises and project work meant to shed light on the exposed theory, the computational project and eventual individual student projects.

The total load is approximately 170 hours, corresponding to **7 ECTS** in Europe. The days were organized as follows: 9.30-13 lectures, time for exercises with assistance (including lunch) till 18-19 (3-4 hours of allocated exercise sessions per day). The course was held at the premises ECT* (Villazzano) in Trento, Italy, from July 3 to July 21 in 2017. Professor Takaharu Otsuka gave a special lecture on Shell-Model Monte Carlo methods in the second week and in addition, five students gave presentations of their own research. These lectures were closely related to the topics of the present Talent course.

All learning material, with preparatory material on second quantization plus Alex Brown's lecture notes were provided well in advance before the school (three weeks) and available via the Github link .

The students were divided into nine groups from the very first day of the school. These groups remained stable throughout the whole duration of the course. Their basic set up consisted in having at least one experimentalist and one theorist in every group. Three groups had four members while the remaining six groups consisted of three members only. This can be summarized in

- We used group work to carry out the tasks.
- Groups were carefully put together to maximize diversity of backgrounds.
- The weekly results and progress of every group was presented at the end of week (Friday afternoon) in a conference-like setting to create accountability.
- We several organized events (weekly social dinners as well) where individuals and groups exchange their experiences, difficulties and successes to foster interaction.
- During the school, on-line and lecture-based training tailored to technical issues were provided. The participants learnt to use and interpret the results of computer-based and hand calculations of nuclear models. The lectures were aligned with the practical computational projects and exercises and the lecturers were always available to help students and work with them during the project and exercise sessions.
- During the second week, after having introduced the basic theory needed to understand full configuration interaction theory (the basis for shell-model studies), the theoretical interpretations were accompanied by talks on how to perform nuclear structure experiments. These lectures were given by Alexandra Gade and Robert Grzywacz and illustrated nicely how nuclear structure theory can be used to interpret data. Several of the experimental students gave additional lectures the same week on additional experimental techniques. Overall, these lectures illustrated nicely, and were highly appreciated by the students, how tightly interwoven theory and experiment are.
- There was enough flexibility and time to organize mini-lectures and discussion sessions on an ad-hoc basis.
- Each group of students maintained an online logbook of their activities and results using Github. This allowed the teaching team to monitor the weekly progress of each group. The weekly presentations were also stored at the Github address of each group. Each group had to establish a Github address where all material was placed.